

FitBuddy – AI Fitness Plan Generator using Gemini Models

Solution Architecture

Project Detail	Value
Project Title	FitBuddy – AI Fitness Plan Generator using Gemini Models
Team ID	SWTID-2026-4989
Team Size	5
Team Leader	Deepika G
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Solution Architecture

Layer	Component	Description	Technology
Presentation	User Interface	Collects user details and shows the plan	Streamlit
Application	Prompt Builder	Creates a prompt from user details	Python
AI	Gemini Model	Generates the workout and diet plan	Google Gemini API
Application	Output Formatter	Formats the plan into a 7-day view	Python
Security	API Key Handling	Keeps the key outside the source code	Environment variables

Flow

Step	Description
1	User enters age, weight, height, goal and fitness level
2	App builds a prompt and sends it to the Gemini API
3	Gemini returns a personalized workout and diet plan
4	App formats the plan and shows it to the user, with a download option